FAULT CURRENT STUDIO - ENGINEERING REPORT
BROWSER EXPORT - ENGINEERING REVIEW REQUIRED

===== SUMMARY =====
Project: Fault Current Studio advanced study
Method: conductor-duty-v28
Model status: BOUNDED_DOCUMENTED_ADIABATIC_CONDUCTOR_I2T_SCREEN
Captured UTC: 2026-10-26T06:08:07.473Z
Warning: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Report data provenance: Entered study inputs, isolated calculated method outputs. No external OEM certification or auto-selected breaker/PPE.

===== STUDY INPUTS =====
Input path: method | Entered value: linear-rms-trace
Input path: studyReference | Entered value: ILLUSTRATIVE conductor duty screening example only
Input path: conductor.areaMm2 | Entered value: 120
Input path: conductor.kASqrtSPerMm2 | Entered value: 100
Input path: conductor.kSourceReference | Entered value: ILLUSTRATIVE k coefficient; NOT OEM qualified; replace before project use
Input path: conductor.material | Entered value: copper
Input path: conductor.initialTempC | Entered value: 90
Input path: conductor.maximumFaultTempC | Entered value: 250
Input path: fault.sourceReference | Entered value: ILLUSTRATIVE 10 kA RMS and 0.2 s; replace with verified clearing envelope
Input path: fault.clearingTimeS | Entered value: 0.2
Input path: fault.samples[0].timeS | Entered value: 0
Input path: fault.samples[0].rmsA | Entered value: 10000
Input path: fault.samples[1].timeS | Entered value: 0.1
Input path: fault.samples[1].rmsA | Entered value: 8000
Input path: fault.samples[2].timeS | Entered value: 0.2
Input path: fault.samples[2].rmsA | Entered value: 4000
Input path: marginFactor | Entered value: 1.1

===== EQUIPMENT DUTY =====
Calculated output path: status | Value: BOUNDED_DOCUMENTED_ADIABATIC_CONDUCTOR_I2T_SCREEN
Calculated output path: scope | Value: One documented conductor, one fault interval, user-qualified adiabatic k-factor, exact integral of entered piecewise-linear RMS current trace or entered protective let-through I^2t. This is a bounded thermal stress screen, NOT an IEC 60949 full calculation, NEC/IEC cable rating, protective coordination, or OEM equipment approval.
Calculated output path: method | Value: linear-rms-trace
Calculated output path: conductorMaterial | Value: copper
Calculated output path: conductorAreaMm2 | Value: 120
Calculated output path: initialConductorTempC | Value: 90
Calculated output path: maximumFaultConductorTempC | Value: 250
Calculated output path: enteredKASqrtSPerMm2 | Value: 100
Calculated output path: kSourceReference | Value: ILLUSTRATIVE k coefficient; NOT OEM qualified; replace before project use
Calculated output path: faultSourceReference | Value: ILLUSTRATIVE 10 kA RMS and 0.2 s; replace with verified clearing envelope
Calculated output path: studyReference | Value: ILLUSTRATIVE conductor duty screening example only
Calculated output path: clearingTimeS | Value: 0.2
Calculated output path: sourceKind | Value: piecewise-linear documented RMS current envelope
Calculated output path: enteredPeakRmsA | Value: 10000
Calculated output path: documentedFaultI2tA2s | Value: 11866666.666666666
Calculated output path: adiabaticCapacityA2s | Value: 144000000
Calculated output path: marginFactor | Value: 1.1
Calculated output path: marginAdjustedDemandA2s | Value: 13053333.333333334

Calculated output path: utilizationPercent | Value: 9.064814814814817
Calculated output path: minimumAreaMm2ForEnteredDemand | Value: 36.12939707957128
Calculated output path: maximumConstantRmsAForClearingTime | Value: 25584.085962673253
Calculated output path: withinEnteredAdiabaticCapacity | Value: true
Calculated output path: limitations[0] | Value: One physical conductor only; multiple parallel runs need independently verified fault current distribution
Calculated output path: limitations[1] | Value: Entered k is NOT derived or verified: material, insulation, initial and maximum temperatures and relevant standard edition must be confirmed by a qualified engineer
Calculated output path: limitations[2] | Value: Current RMS trace is assumed piecewise-linear between samples; waveform/DC offset/limiting effects are NOT inferred
Calculated output path: limitations[3] | Value: Documented I^2t must be for the same physical conductor and entire clearing interval; device trip time and let-through are NOT calculated
Calculated output path: limitations[4] | Value: Adiabatic only: no non-adiabatic corrections, adjacent heating, cable joints/terminations or sheath/armour stress
Calculated output path: limitations[5] | Value: Passing this screen is not cable sizing, installation compliance, or protective-device approval

===== ASSUMPTIONS & QA =====
Evidence / limitation: Release status | Value: BOUNDED METHOD, NOT A STANDARDS/DEVICE APPROVAL
Evidence / limitation: Warning | Value: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Evidence / limitation: Study method | Value: conductor-duty-v28
Evidence / limitation: Timestamp | Value: 2026-10-26T06:08:07.473Z
Evidence / limitation: outputs.scope | Value: One documented conductor, one fault interval, user-qualified adiabatic k-factor, exact integral of entered piecewise-linear RMS current trace or entered protective let-through I^2t. This is a bounded thermal stress screen, NOT an IEC 60949 full calculation, NEC/IEC cable rating, protective coordination, or OEM equipment approval.
Evidence / limitation: outputs.limitations[0] | Value: One physical conductor only; multiple parallel runs need independently verified fault current distribution
Evidence / limitation: outputs.limitations[1] | Value: Entered k is NOT derived or verified: material, insulation, initial and maximum temperatures and relevant standard edition must be confirmed by a qualified engineer
Evidence / limitation: outputs.limitations[2] | Value: Current RMS trace is assumed piecewise-linear between samples; waveform/DC offset/limiting effects are NOT inferred
Evidence / limitation: outputs.limitations[3] | Value: Documented I^2t must be for the same physical conductor and entire clearing interval; device trip time and let-through are NOT calculated
Evidence / limitation: outputs.limitations[4] | Value: Adiabatic only: no non-adiabatic corrections, adjacent heating, cable joints/terminations or sheath/armour stress
Evidence / limitation: outputs.limitations[5] | Value: Passing this screen is not cable sizing, installation compliance, or protective-device approval